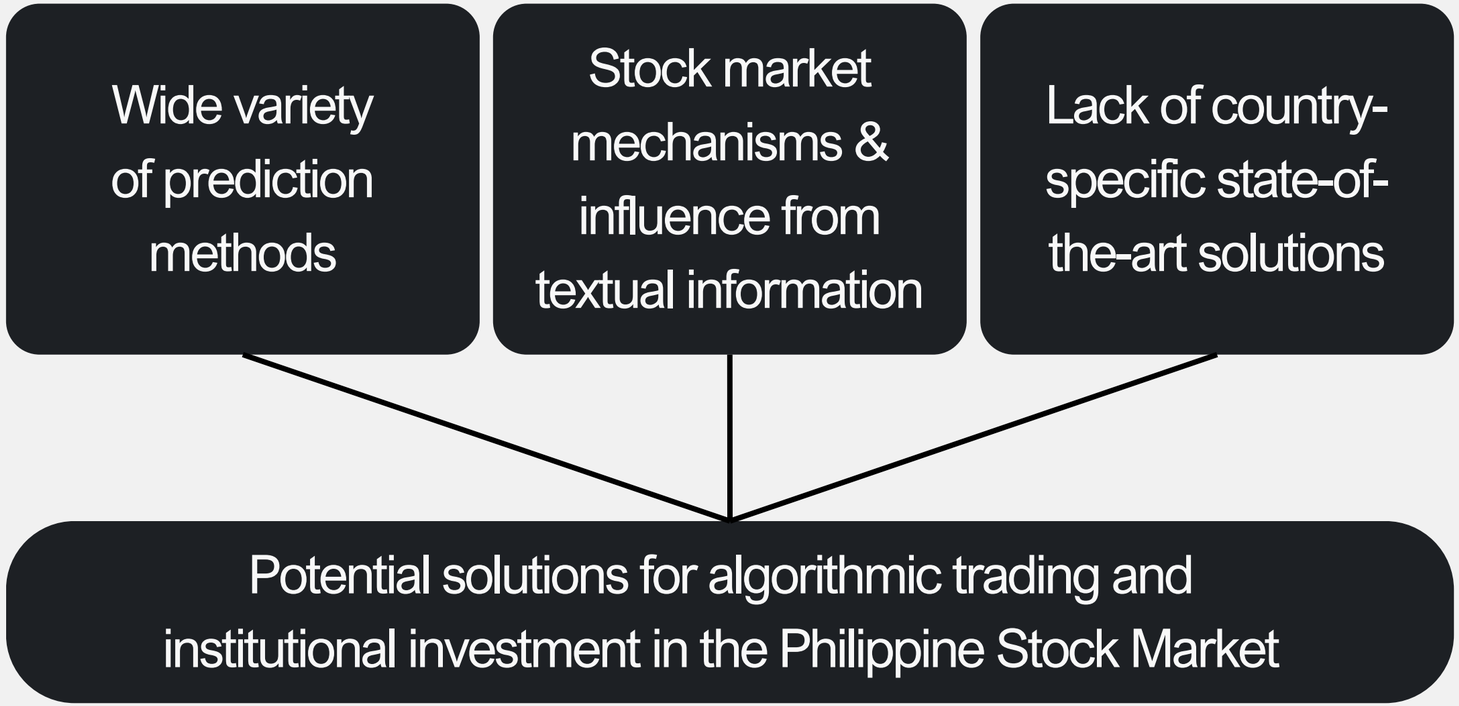
progress report

Evaluating NLP and Deep Learning Approaches for Global Forecasting of Intraday Philippine Stock Market Reactions Using Heterogeneous Data Sources

presented by
Marcus Tornea

Background of the Study



Background of the Study

1. Lack of multimodal intraday literature
2. Lack of modality comparison studies
3. Lack of studies taking into account time-scale differences in market behavior
4. Lack of complex explainability analyses including attention mechanisms

Statement of the Problem

What characteristics of a deep learning model leads to better predictions?

Research Objectives

1. Which models offer an overall improvement over statistical and machine learning baselines?

2. How does each one of prediction horizon, inclusion of news, and inclusion of social media posts affect model performance, both directly and indirectly through joint interaction effects?

3. How does model performance evolve over time, and how does each one of prediction horizon, inclusion of news data, and inclusion of social media interact with this evolution, both directly and indirectly through joint interaction effects?

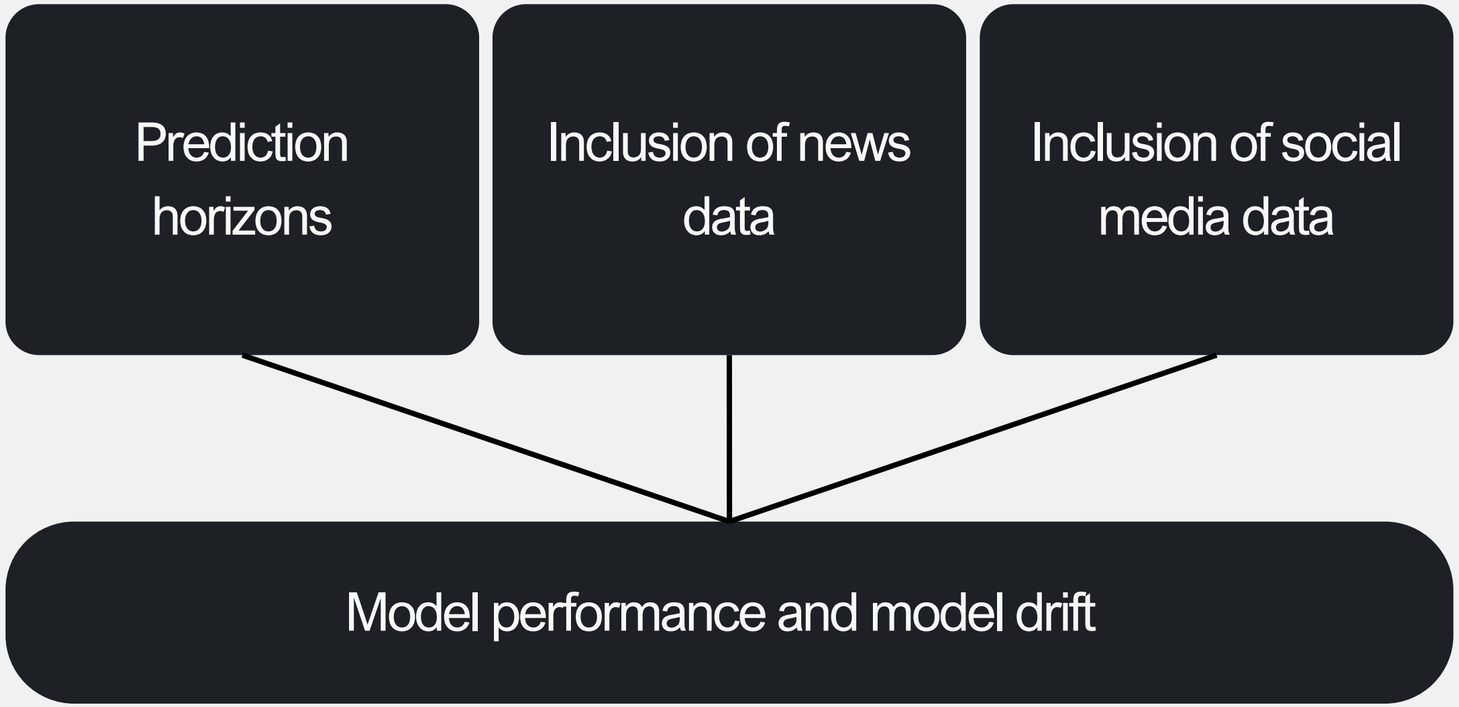
4. Why do prediction horizon, inclusion of news data, and inclusion of social media data complement or interfere with each other's effects on model performance?

5. Why does each one of the prediction horizon, inclusion of news data, and inclusion of social media interact with or not affect model drift?

Expected Analytics Results

- **Factorial analysis results for model performance and drift**
  - Quantification of main and interaction effects of the independent variables, with statistical significance
  - Difference in model performances between the best experimental setting and corresponding heuristic/statistical/machine learning methods
- **Exploratory data analysis of explainability mechanisms**
  - SHAP/LIME values for feature groups (technical data, news data, social media data)
    - Evolution of values over time
    - Differences in explainability mechanisms between experimental settings

Methodology



Two major parts:

- Factorial analysis
- Interpretability analysis

Experimental Setup

Hard-parameter Sharing Model

All predicted targets have a shared market/news/social media representation, but each one has its own prediction head.

Data Treatment

Generating scores

- For each classification target:
  - Accuracy score per stock for each setting

Statistical tests

- Random intercept mixed-effects model for main and interaction effects
- Wilcoxon signed-rank test for best model comparison with other baselines

Testing

Trading Simulations

- For high-frequency retail investment without trading fees.
- For institutional investment:
  - Example: predictions shall be used to moderate frequency of trading, instead of dictating decisions to trade or not.

Interpretability Analysis

Comparing differences in SHAP/LIME values across prediction horizons, between data modalities

- SHAP values diminishing more quickly for a certain modality would mean that the market absorbs that information modality at a faster rate.

Evolution of SHAP values over time

- SHAP values changing over short periods of time may uncover model drift mechanisms.

Accomplishments

Current Progress

- Data Collection
- Model Design
- Training
- Methodology Revisions

Accomplishments

- **Collected Data:**
  - All 30 blue chip stocks
  - All country and sector indices
  - US to PHP exchange rates
  - Gold to PHP exchange rates
  - Copper futures
  - Oil futures
  - 10-year government bonds

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  - Currently finished collecting: 25,000 news articles
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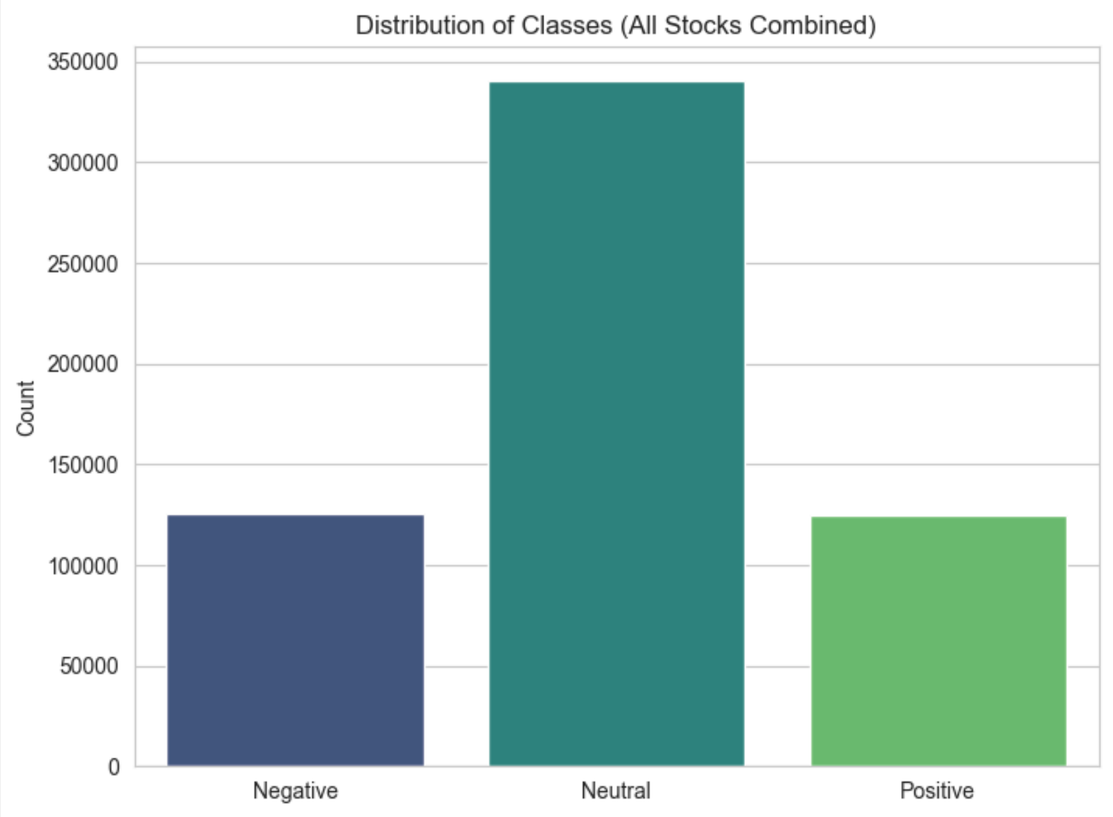
  - Purchased credits for collecting 60,000-70,000 X posts
  - Will collect 25,000 more, to reach a total of 50,000 news articles
  - Will purchase more X scraper API credits to reach 120,000-140,000 X posts

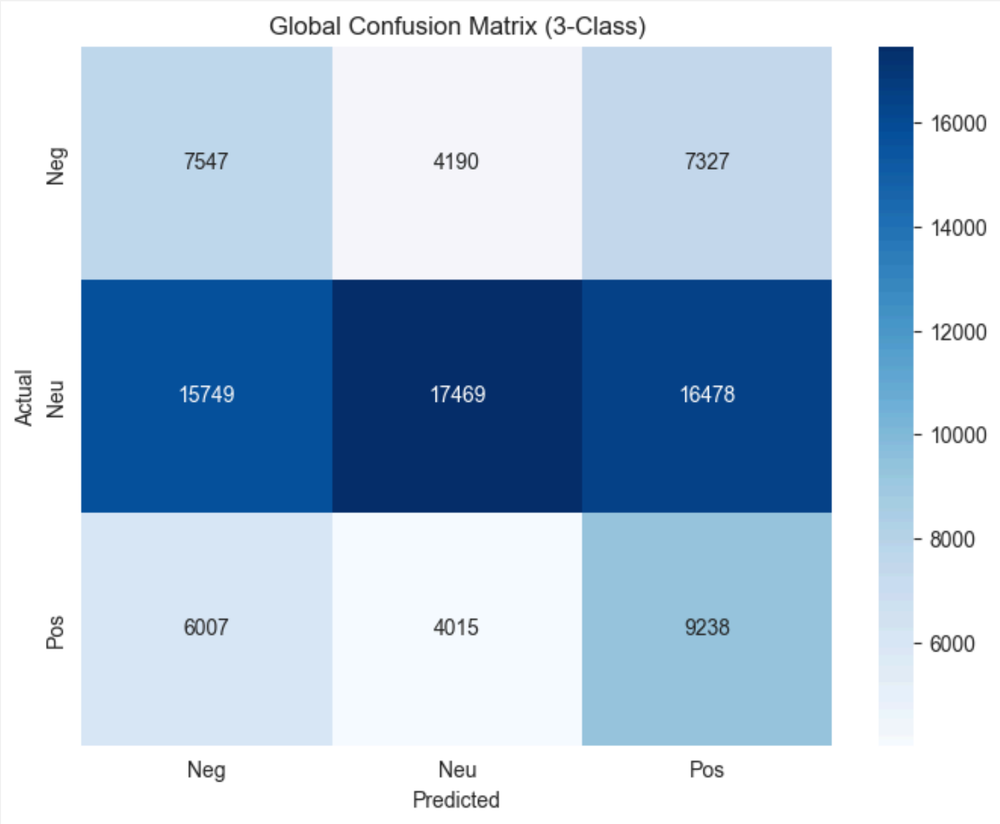
Accomplishments

- **Stacked LSTM model:**
  - Two layers (128 units and 64 units)
  - 60-minute lookback period
  - Two dense layers (27-dim and 3-dim outputs)

Accomplishments

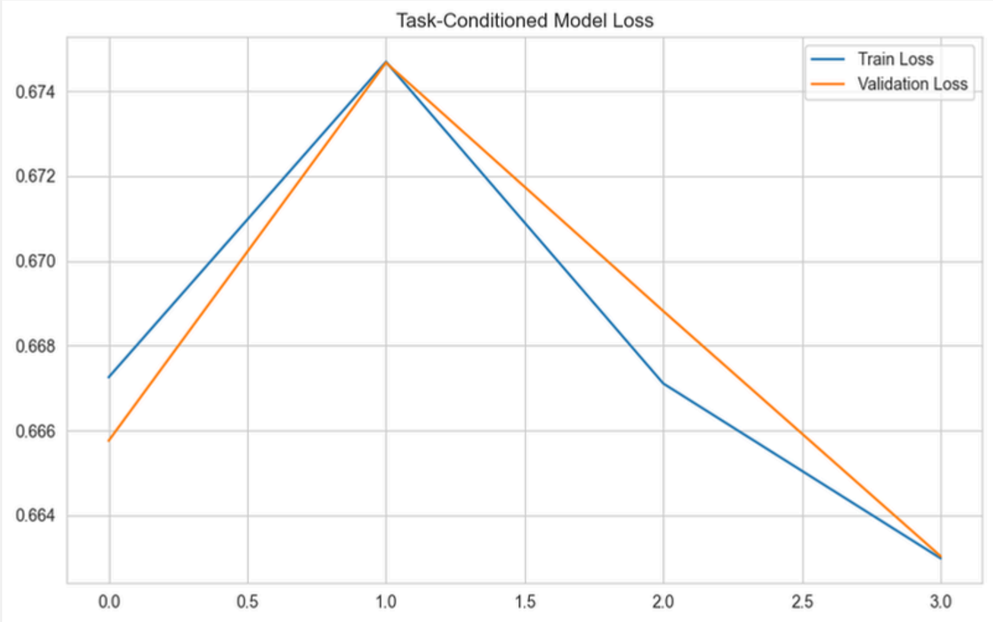
Model accuracy: 39%; F1 score: negative: 31%, neutral: 46%, positive: 35%





Accomplishments

- **Experimental Regression LSTM:**
  - Two layers (128 units and 64 units), 60 units, single dense layer output



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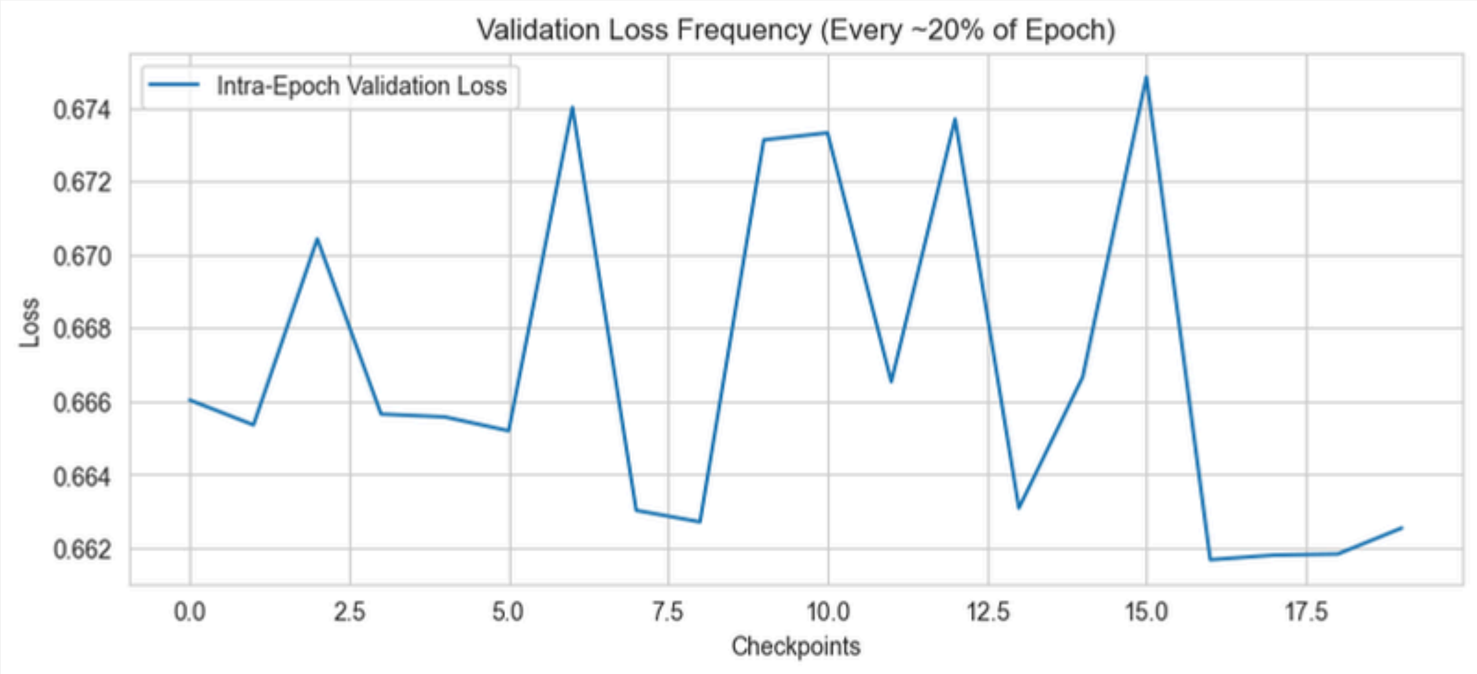

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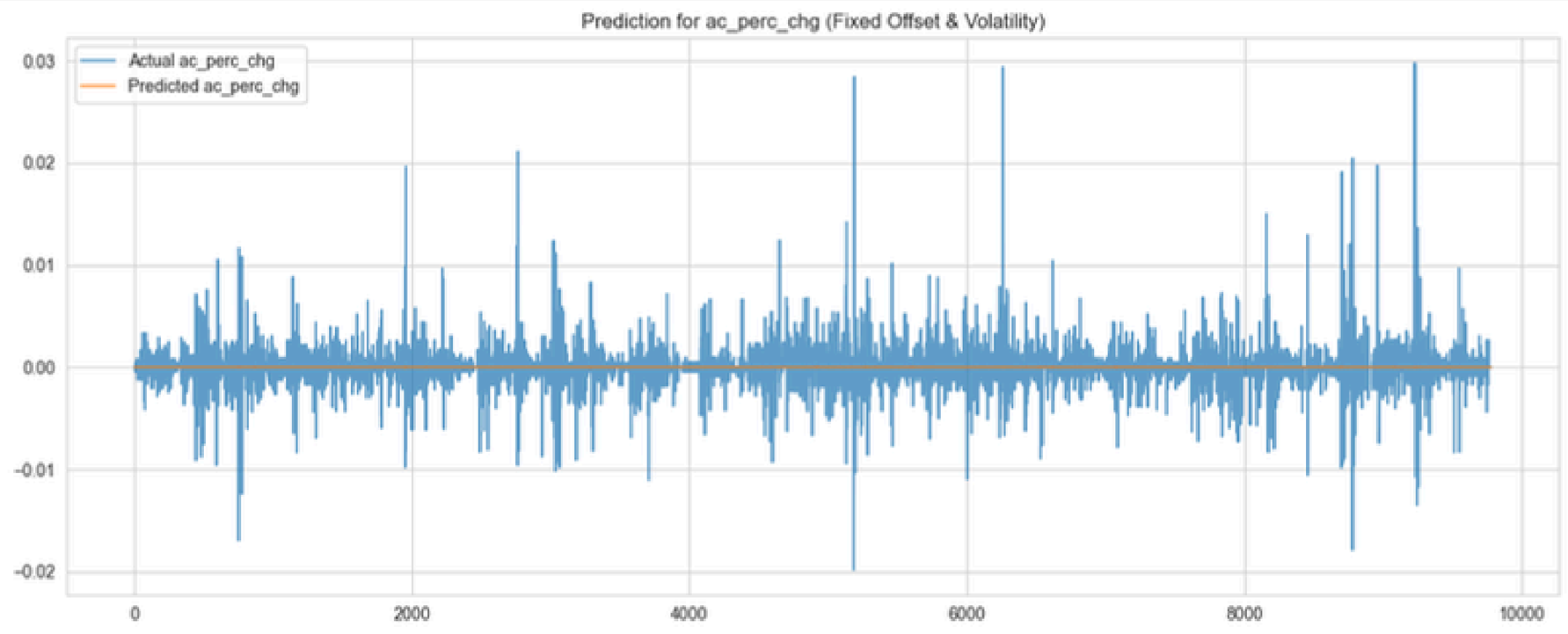

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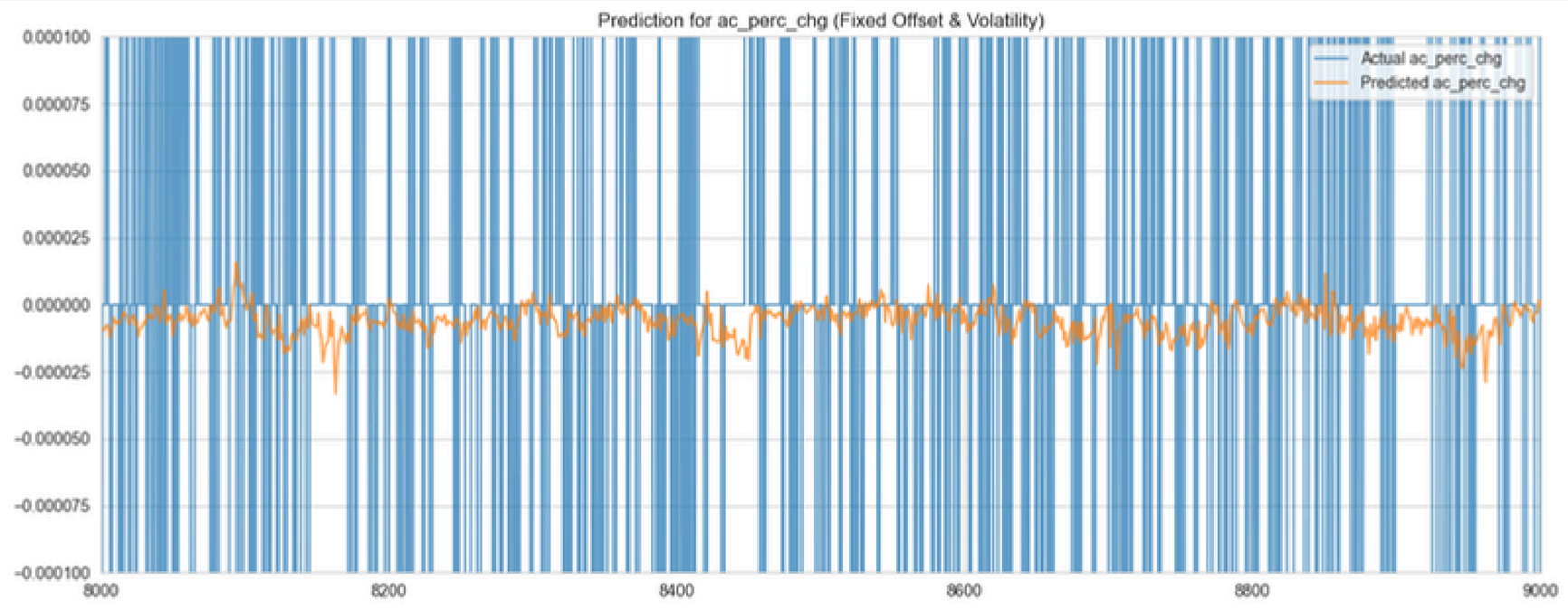

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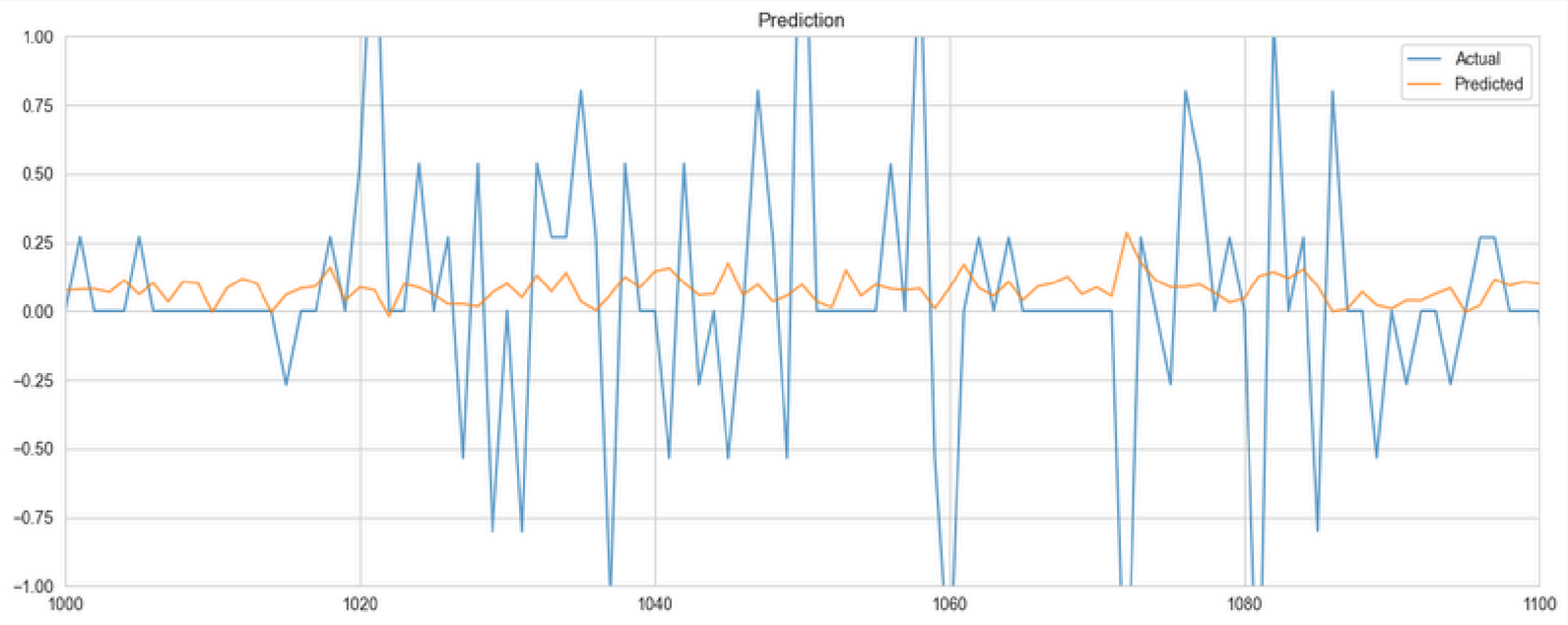

Remaining Workplan

- **New workflow:**
  - Instead of just one model predicting every stock, train one model per stock

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  - **Technical data feature engineering → feature selection**

Remaining Workplan

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  - Trend-based: MACD, ADX (and accompanying directional indicators)
  - Volatility: Bollinger bands, average true range
  - Volume-based: relative volume, on-balance volume

Remaining Workplan

- **Model performance measures:**
  - Accuracy scores for **intrinsic** model performance
  - Sharpe ratios (given a “good” intraday retail trading strategy)—dependent on trading strategies

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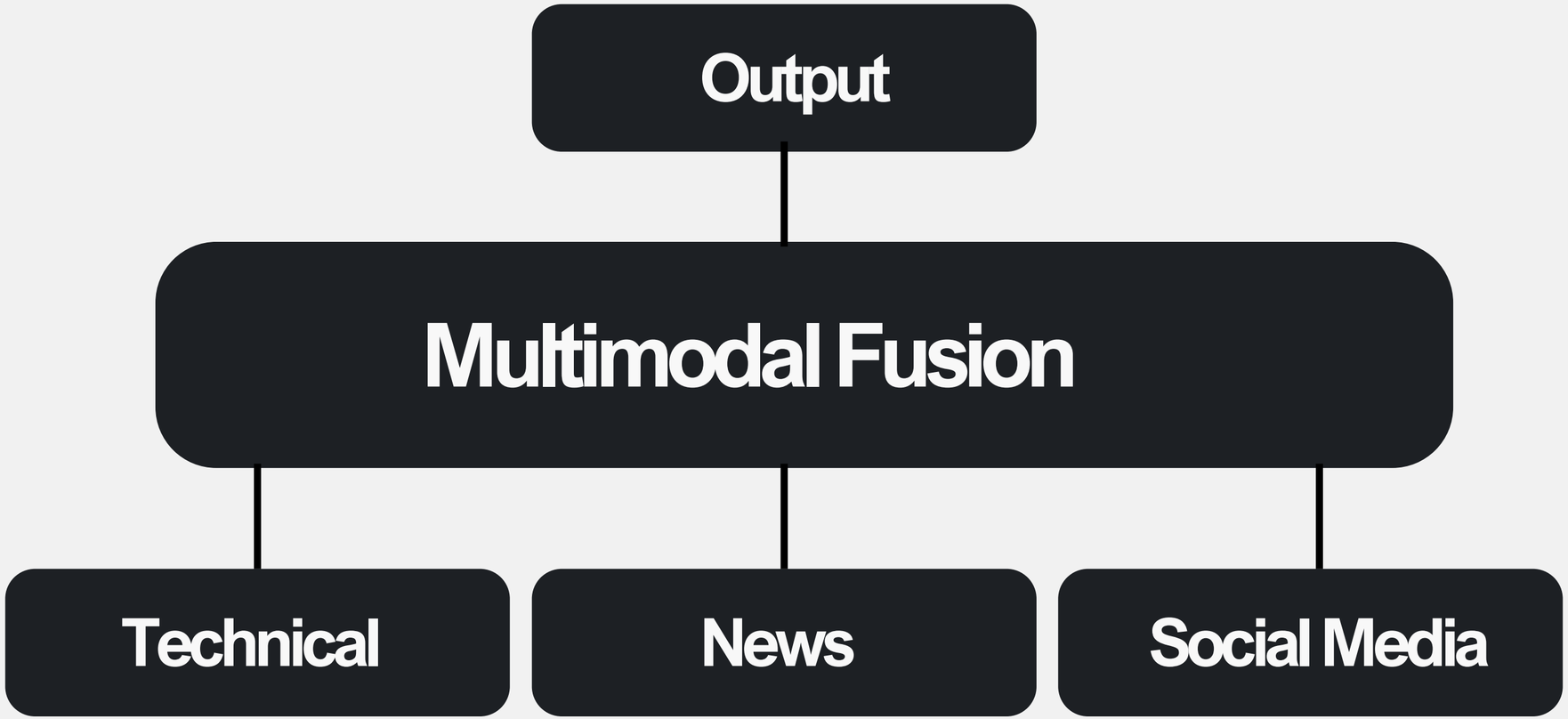
- **Model drift scores:**
  - ADWIN-based scoring strategy
    - Makes use of ADWIN-generated windows; scores are computed by difference between model errors and window means

Remaining Workplan

- **To optimize news and social media processing:**
  - Only contextual vectors for news headlines and X posts will be taken; no more hierarchical attention processing
  - Cross attention will be used so that article/post sequences can be processed as a sequential input, not as a pooled vector

Remaining Workplan



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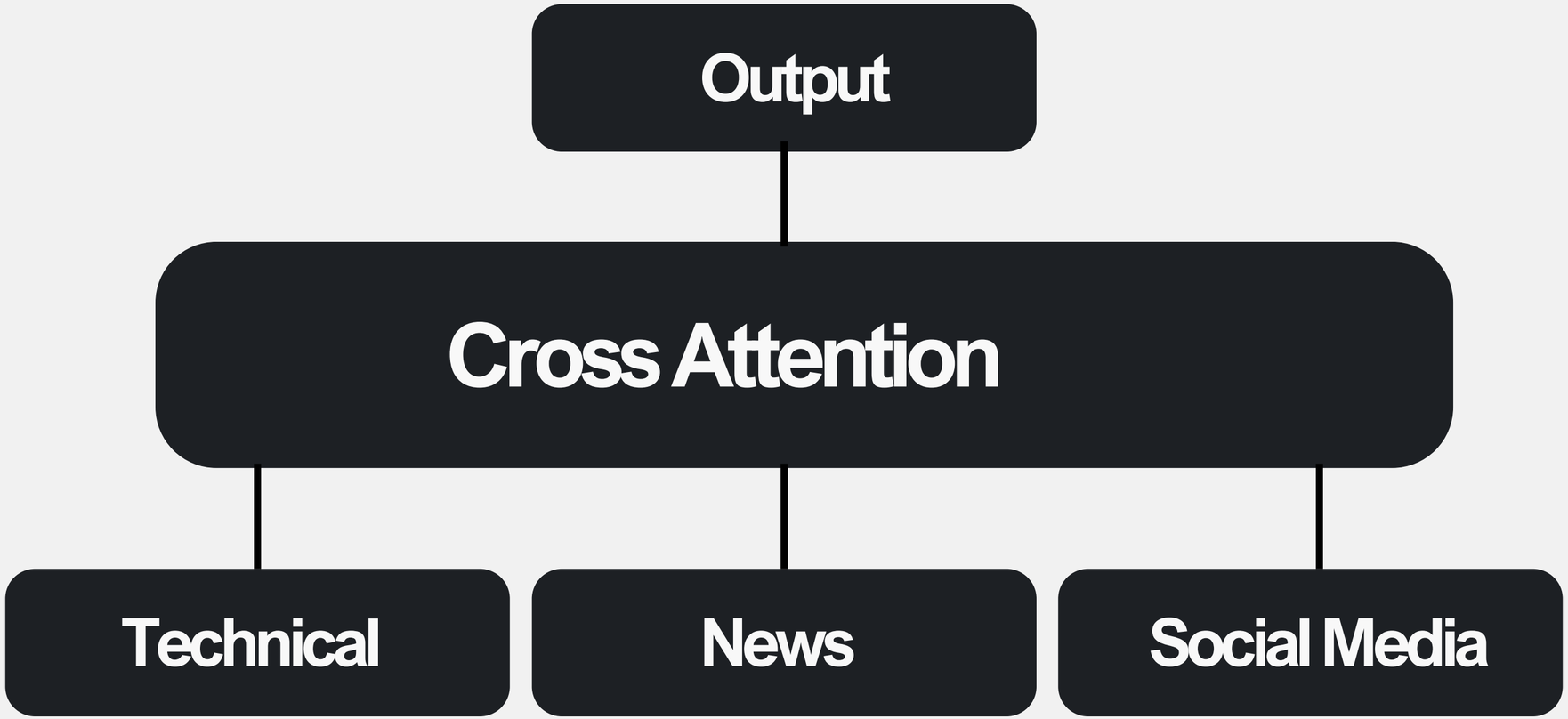

Remaining Workplan

- Training
- Testing and trading simulations
- Interpretability Analysis
- Write-up

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